

Roshan Raj

Curriculum Vitae

Upto May 2024

Personal Details

Name: Roshan Raj
Nationality: Indian
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Research interests

Reconstruction and Foundations of Quantum Theory, Quantum Field Theory, Quantum to Classical Transition, Gravity and EM Unification, Mathematical Physics. (See: List of problems I resonate with! [Get the PDF])

Educational Details

2019-24: 5 years Integrated M.Sc. (Physics), Sardar Vallabhbhai National Institute of Technology, Surat, India. CGPA: **9.33/10** (IX sem)

Master's Dissertation

Title: *Quantum and Classical Applications of Relativistic Relationship of Kinetic Energy and Momentum*

Supervisor: Dr Vikash K Ojha, SVNIT-Surat; [Jul, 2023-May, 2024]

Brief Abstract: A newly found classical relation between relativistic kinetic energy K and relativistic momentum $\mathbf{p}$, namely $|\mathbf{p}|^2 = (1 + \gamma)mK$, involving Lorentz factor γ , leads to a compact relativistic quantum wave equation $\mathbf{p}^\mu \partial_\mu \Psi = 0$, Where $\mathbf{p}^\mu = [(1+\gamma)p_0, i\hbar\gamma\partial_j]$ with Minkowski signature as $(+, -, -, -)$ is modified momentum four-vector. The reported relation is replacing both the Newtonian non-relativistic energy-momentum formula and relativistic total energy-momentum relation, $E^2 = p^2c^2 + m^2c^4$, and that has sought application in both quantum and classical relativistic mechanics. Its application predicted the contractions of Bohr radius at relativistic speeds, corrected the expression of the Relativistic Lagrangian of a free particle, etc.

Institute Mini-projects

- Title: New light on the concepts of Observable and indeterminacy in the quantum realm
Guide: Dr Vikash K Ojha, SVNIT-Surat– UG Project III, [Jul-Dec, 2022]–URL: <https://shorturl.at/6lMIS>
Target area: Quantum to Classical Transition
- Title: On the investigation of two non-neutral static bodies
Guide: Dr Sharad K Yadav, SVNIT-Surat– UG Project II, [Jan-May, 2022]– DOI: [10.48550/arXiv.2209.10641](https://doi.org/10.48550/arXiv.2209.10641)
Target area: Gravity and EM Unification
- Title: Dynamics of Two Objects Considering the Minimum Total Potential Energy Principle
Guide: Dr. Himanshu Pandey, SVNIT-Surat– UG Project I, [Jul-Dec, 2021]–DOI: [10.31219/osf.io/ukx9t](https://doi.org/10.31219/osf.io/ukx9t)
Target area: Gravity and EM Unification

Independent Works

- Raj, Roshan. 2023. “Obtaining the Klein-Gordon Wave Equation Without Using the Quantum Operators.” OSF Preprints. Oct 30. doi:10.31219/osf.io/6hmyf.
Target area: Reconstruction and Foundations of Quantum Theory

- Raj, Roshan. 2023. “Reviewing Observables in Classical and Quantum Mechanics.” OSF Preprints. Oct 10. doi:10.31219/osf.io/p2ufx.
Target area: Foundations of Quantum Theory
- Raj, Roshan. 2023. “An Open-ended Story on Quantization.” OSF Preprints. Oct 8. doi:10.31219/osf.io/92bvp.
Target area: Foundations of Quantum Theory

Internship

- Title: Pramatrik Gatiki ke Chitra [*The pictures of quantum dynamics* (Hindi)]
Guide: Dr Vikash K Ojha, SVNIT-Surat – Internship I, [May-Jul, 2022]–URL: <https://shorturl.at/lvEIE>
Target area: Physics education for Hindi Readers.

Achievements

- 09/16-present: NTS Scholarship of merit, Government of India, New Delhi.
02/20: Winner (Physics), InQuest 4.0, SCOSH [Student Chapter], SVNIT

Soft Skills

- Programme: C, Python, MATLAB, Octave
Softwares: L^AT_EX, MS Excel, Mathematica, and use of GitHub.
Web Design: Basics of HTML5, CSS3
Languages: English (Intermediate), Hindi(Native)

Workshops & Webinars

- 07/21: *Quantum Fields, Geometry & Representation Theory*, ICTS-TIFR, Bangalore.
12/20: National workshop on *Data analysis using MS Excel*, BBD NITM, Lucknow.
08/20: National workshop on *MATLAB Tools And Applications*, BBD NITM, Lucknow.

Books & Talks

- 02/07/23: MaNoVighn: Short Poetry Collections, Kindle, ASIN: B0C9Y7X1PG
01/09/21: KālSanGharsha: Short Poetry Collections, Kindle, ASIN: B09FC1XRPK
13/11/21: *Unification in Mathematics*, IntERAct Seminar series, AMHD, SVNIT-Surat

Responsibilities & Positions

- 11/22-01/23: **Design-Infra Coordination Head & Member:** *Manoj Memorial Night Cricket Tournament* (MMNCT) Annual cricket event, SVNIT Surat
10/21-07/22: **Joint Academic Affairs Secretary:** *Academic Affairs Council* SVNIT, Surat
06/20-07/22: **Co-convenor (03/21) & Member** *Society for Cultivation of Science and Humanities* (SCOSH) Student Chapter, SVNIT Surat

I have always been an enthusiastic, independent seeker of truth. I will listen to mathematical beauty, ancient history, and philosophical ideas and stand up for social justice. My hobbies include Hindi Poetry, Graphic design, and writing articles.